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Center for Subsurface Energy and Environment
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Research Focus

My research focuses on sustainable energy infrastructure, specifically on coupled mechanics, fluid and heat transport, and fracture evolution in natural and engineered geologic systems. I develop approaches that enable reliable and resilient geotherm energy, geologic carbon storage, and long-duration underground energy storage.

Academic Appointments

<i>The University of Texas at Austin</i>	Austin, TX
Postdoctoral Fellow, Center for Subsurface Energy and Environment	08/2026 – Present
<i>The Pennsylvania State University</i>	University Park, PA
Postdoctoral Scholar, EMS Energy Institute	08/2024 – 07/2026

Education

<i>The Pennsylvania State University</i>	University Park, PA
Ph.D. in Energy and Mineral Engineering, College of Earth and Mineral Sciences	01/2021 – 08/2024
Ph.D. Minor in Computational Science, College of Engineering	11/2022 – 08/2024
<i>University of Science and Technology Beijing</i>	Beijing, China
M.Eng. in Fluid Mechanics, College of Civil and Resource Engineering	09/2017 – 06/2020
<i>University of Science and Technology Beijing</i>	Beijing, China
B.Eng. in Building Environment and Energy Engineering Application	09/2013 – 06/2017

Selected Research Experience

1. High-Efficiency and Resilient Enhanced Geothermal Systems 12/2021 – Present

Develop temperature-responsive proppants and thermally activated cellular screens (TACS) to actively regulate fracture-scale flow, mitigate thermal short-circuiting, and extend the operational lifetime of enhanced geothermal systems (EGS). These technologies are designed and evaluated using coupled thermo-hydro-mechanical-wellbore (THMW) modeling to resolve interactions among fracture evolution, heat transport, fluid flow, and wellbore conditions.

2. Repurposing Existing Infrastructure for Clean Energy 12/2023 – Present

Investigate the repurposing of existing infrastructure, including abandoned oil and gas wells, underground mines, and associated mine-water systems, for geothermal heat extraction, long-duration energy storage, and low-carbon data-center cooling.

3. Carbon Management for Sustainable Energy Infrastructure 04/2025 – Present

Develop physics-based and data-driven approaches to improve prediction, monitoring, and risk assessment for geologic CO₂ storage in heterogeneous reservoir systems. This work combines physics-informed machine learning, coupled subsurface-flow

modeling, and geomechanical analysis to characterize CO₂ plume evolution, uncertainty, containment performance, and fault-reactivation risk, supporting the long-term decarbonization of energy and industrial systems.

Teaching Experience

Guest Lecturer, Department of Energy and Mineral Engineering, Penn State 08/2024 – 12/2025

- PNG 475/577, Production and Completions Engineering (Fall 2024 and 2025): Delivered lectures on inflow performance relationships (IPR), near-wellbore skin, and hydraulic fracture mechanics.
- PNG 440W, Formation Evaluation (Fall 2024): Delivered instruction on quantitative interpretation of subsurface measurements.
- EGEE 101H, Energy and the Environment (Spring 2025): Served as a guest instructor for one class session; delivered instruction on natural gas and nuclear energy and evaluated two student team presentations.

Teaching Assistant, Department of Energy and Mineral Engineering, Penn State 08/2021 – 05/2024

- PNG 475, Production and Completions Engineering, Spring 2022, 2023, and 2024
- PNG 440W, Formation Evaluation, Fall 2023
- EGEE 494, Energy and the Environment Research Project, Fall 2021

Undergraduate Research Co-Mentor, Penn State Summers 2022 and 2023

- Co-mentored two undergraduate students with Prof. Arash Dahi Taleghani on hydraulic fracturing research; provided day-to-day guidance on model development, data analysis, technical writing, and final presentations.

Proposal Development Experience

Funded Proposals

1. **Autonomous and Tunable Smart Screens to Control Thermal Evolution in EGS Wells**
DOE ARPA-E, 2024 | **Role:** Key Personnel and Primary Narrative Writer | **Status:** Funded
2. **Efficient, Low-cost, and Environmentally Benign Production of Lithium Battery Precursors from Geothermal Resources**
DOE EERE-AMMTO, 2023 | **Role:** Proposal Contributor | **Status:** Funded
3. **Temperature-Sensitive Hydraulic Conductivity Controller Proppants for Enhanced Geothermal Systems**
DOE EERE-GTO, 2022 | **Role:** Key Personnel and Proposal Contributor | **Status:** Funded

Submitted Proposals

1. **Autonomous Fracture Regulation: Deep Learning-Based Fracture Control for High-Efficiency EGS**
DOE Genesis Mission, 2026 | **Role:** Key Personnel; Lead Writer for the Technical Narrative | **Status:** Submitted
2. **Physics-Synergized Deep Learning Framework for High-Fidelity Spatiotemporal Evolution of Geological CO₂ Storage**
DOE Genesis Mission, 2026 | **Role:** Key Personnel; Lead Writer for the Technical Narrative | **Status:** Submitted
3. **Enhanced Heat Transfer for Closed-Loop Superhot Geothermal Systems Through Advanced Heat Pipe Technology**
DOE ARPA-E, 2026 | **Role:** Key Personnel; Lead Writer for the Technical Narrative | **Status:** Submitted
4. **Standardized CFD Modeling for LNG Shroud Leak Mitigation and Structure Design**
DOT PHMSA, 2025 | **Role:** Key Personnel; Lead Writer for the Technical Narrative | **Status:** Submitted

5. Electromagnetic Heating for Sustainable Gas Hydrate Extraction
ACS PRF, 2024 | **Role:** Key Personnel; Lead Writer for the Technical Narrative | **Status:** Submitted

Awards and Honors

EMS Postdoctoral Excellence in Research Award, Penn State (\$1,000) ↗	2026
Penn State Postdoc Travel Award (\$500)	2026
Outstanding Reviewer Award 2024-2025, Int. J. Coal Sci. Tech.	2025
EME Student Merit Award, Penn State (\$3,000)	2024
Chinese Government Award for Outstanding Self-Financed Students Abroad	2023
National Scholarship of China	2019
First-Class Academic Scholarship, University of Science and Technology Beijing	2018
Second Prize, National Petroleum Engineering Design Competition, China	2018

Peer-reviewed Journal Articles

1. **Zhang, Q.**, Dahi Taleghani, A[✉], Zhan, Z., Li, G. Enhanced heat recovery in CO₂-enhanced geothermal systems: A temperature-sensitive proppant approach. *Applied Thermal Engineering*, 2026, 129701.
2. **Zhang, Q.**, Dahi Taleghani, A[✉], Ji, K., Hua, W. Coupled Thermo-Hydro-Mechanical-Wellbore Modeling of Thermal Short-Circuiting in Enhanced Geothermal Systems. *Energy*, 2026, 140038.
3. Du, S.[✉], Wang, S., **Zhang, Q.** High-fidelity spatiotemporal evolution of CO₂ storage in saline aquifers: A physics-synergized deep learning framework for CO₂ plume prediction and super-resolution reconstruction. *International Journal of Coal Science & Technology*, 2026, 13(1), 90.
4. Du, S.[✉], Wang, S., **Zhang, Q.** Physics-Informed Hybrid Transformer with Bayesian Optimization for Intelligent Geological CO₂ Sequestration Modeling. *Energy & Fuels*, 2026, 40(23), 12394-12410.
5. Hua, W., **Zhang, Q.**, Dahi Taleghani, A[✉]. Repurposing abandoned oil and gas wells for sustainable greenhouse heating: a geothermal energy approach. *Applied Thermal Engineering*, 2026, 297, 130824.
6. Zhan, Z., **Zhang, Q.**, Yu, C., Li, G., Dahi Taleghani[✉], A. Reversible actuation of artificial muscle proppants for self-regulating fracture flow and long-term geothermal sustainability. *Renewable Energy*. 2026, 126512.
7. **Zhang, Q.**, Dahi Taleghani, A[✉], Elsworth, D. Underground Energy Storage Using Abandoned Oil & Gas Wells and Assisted by Geothermal. *Journal of Energy Storage*, 2025, 115317.
8. **Zhang, Q.**, Dahi Taleghani, A[✉]. Real-Time Temperature Monitoring and Flow Control for Improved Heat Extraction in Enhanced Geothermal Systems. *Energy*, 2025, 134274.
9. Wang, D.[✉], Dong, Y., Wei, C., **Zhang, Q.**, Sheng, M., Yu, B. Expansion-induced fracture propagation in deep geothermal reservoirs under alternate-temperature loading. *Advances in Geo-Energy Research*, 2025, 15(3), 261-272.
10. Ji, K., **Zhang, Q.**, Dahi Taleghani, A., Li, H. Thermal short-circuit mitigation in enhanced geothermal systems: a novel dynamic flow regulation technology using temperature-sensitive viscosity modifiers. *Energy Conversion and Management*, 2025, 343, 120196.
11. Ji, K., Dahi Taleghani, A., Liu, S., **Zhang, Q.**, Dai, K., Li, H., Tang, C. A[✉]. Comprehensive investigation on the heat extraction performance of a novel enhanced geothermal system based on mining technology. *Renewable Energy*, 2025, 249, 123233.
12. **Zhang, Q.**, Dahi Taleghani, A[✉]. Downhole Flow Control: A Key for Developing Enhanced Geothermal Systems in Horizontal Wells. *Renewable Energy*, 2024, 121578.

13. **Zhang, Q.**, Dahi Taleghani, A[✉]. Intelligent fluid flow management for enhanced geothermal system in fractured horizontal wells. *Applied Thermal Engineering*, 2024, 239, 122191.
14. Al Balushi F, **Zhang Q.**, Dahi Taleghani, A[✉]. On the Impact of Proppants Shape, Size Distribution, and Friction on Adaptive Fracture Conductivity in EGS. *Geoenergy Science and Engineering*, 2024, 213115.
15. **Zhang, Q.**, Dahi Taleghani, A. [✉], Li, G. Fracture conductivity management to improve heat extraction in enhanced geothermal systems. *International Journal of Heat and Mass Transfer*, 2024, 218, 124725.
16. Al Balushi, F., **Zhang, Q.**, Dahi Taleghani, A [✉]. Improving enhanced geothermal systems performance using adaptive fracture conductivity. *Applied Thermal Engineering*, 2023, 233, 121206.
17. **Zhang, Q.**, Dahi Taleghani, A. Autonomous fracture flow tuning to enhance efficiency of fractured geothermal systems. *Energy*, 2023, 128163.
18. Al Balushi, F., **Zhang, Q.**, Dahi Taleghani, A [✉]. Autonomous Fracture Conductivity Using Expandable Proppants in Enhanced Geothermal Systems. *SPE Journal*, 2023, 1-15.
19. Yue, M., Zhu, W. Y., Gou, F. F., Song, T. R., You, Y. C., **Zhang, Q**[✉]. Impacts of proppant distribution on development of tight oil reservoirs with threshold pressure gradient. *Petroleum Science*, 2023, 21(1), 445-457.
20. **Zhang, Q.**, Liu, W. [✉], Dahi Taleghani, A. Numerical study on non-Newtonian Bingham fluid flow in development of heavy oil reservoirs using radiofrequency heating method. *Energy*, 2022, 239, 122385.
21. **Zhang, Q.**, Liu, W. [✉], Wei, J., Dahi Taleghani, A. Numerical Simulation Study on Temporary Well Shut-in Method in Development of Shale Oil Reservoir. *Energies*, 2022, 15(23), 9161.
22. **Zhang, Q.**, Dahi Taleghani, A[✉]. On the role of proppants and geomechanics on flowback behavior in complex fracture networks. *Journal of Petroleum Science and Engineering*, 2022, 216: 110835.
23. Liu, W. [✉], **Zhang, Q.**, Dong, Y., Chen, Z., Duan, Y., Sun, H., Yan, X. Analytical and numerical studies on a moving boundary problem of non-Newtonian Bingham fluid flow in fractal porous media. *Physics of Fluids*, 2022, 34(2), 023101.
24. Liu, W. [✉], Duan, Y., **Zhang, Q.**, Chen, Z., Yan, X., Sun, H., Dahi-Taleghani, A. Analytical study on a one-dimensional model coupling both Darcy flow and low-velocity non-Darcy flow with start-up pressure gradient in heterogeneous composite reservoirs. *Journal of Porous Media*, 2022, 25 (7).
25. Zhu, W., Liu, Y., Shi, Y., Zou, G., **Zhang, Q.**, Kong, D[✉]. Effect of dynamic threshold pressure gradient on production performance in water-bearing tight gas reservoir. *Advances in Geo-Energy Research*, 2022, 6(4), 286-295.
26. **Zhang, Q.**, Zhu, W. [✉], Liu, W., Yue, M., Song, H. Numerical simulation of fractured vertical well in low-permeable oil reservoir with proppant distribution in hydraulic fracture. *Journal of Petroleum Science and Engineering*, 2020, 195, 107587.
27. Zhu, W. [✉], **Zhang, Q.**, Yue, M., Zhang, L. Effect of uneven distribution of proppant in fracture network on exploitation dynamic characteristics. *Chinese Journal of Engineering*, 2020, 42 (10), 1318-1324.
28. Yue, M., **Zhang, Q.** [✉], Zhu, W., Zhang, L., Song, H., Li, J. Effect of proppant distribution in fracture networks on horizontal well performance. *Journal of Petroleum Science and Engineering*, 2020, 187, 106816.
29. **Zhang, Q.**, Wei, C., Wang, Y., Du, S., Zhou, Y., Song, H[✉]. Potential for Prediction of Water Saturation Distribution in Reservoirs Utilizing Machine Learning Methods. *Energies*, 2019, 12, 3597.

([✉] represents the corresponding author)

Manuscripts Under Review

1. **Zhang, Q.**, Dahi Taleghani, A. Temperature-Responsive Proppants for Thermal Short-Circuiting Mitigation in Enhanced Geothermal Systems ([Under Review - R3, Energy](#))
2. **Zhang, Q.**, Dahi Taleghani, A., Elsworth, D. Thermal Short-Circuiting in Enhanced Geothermal Systems: Mechanisms, Field Evidence and Emerging Mitigation Technologies ([Under Review - R3, Renewable and Sustainable Energy Reviews](#))
3. **Zhang, Q.**, Liang, Y., Dahi Taleghani, A., Salt-Creep and Depletion-Induced Stress Reorientation: Implications for Fault Slip Tendency in a Deepwater Reservoir ([Under Review, Rock Mechanics and Rock Engineering](#))
4. Ji, K., **Zhang, Q.**, Dahi Taleghani, A. Adaptive fracture-conductivity regulation for thermal short-circuit mitigation in enhanced geothermal system ([Under Review, Physics of Fluids](#))

Conference Presentations with Proceedings

1. **Zhang, Q.**, Dahi Taleghani, A. ICD-Enabled Stress Management to Mitigate Thermal Short-Circuiting in EGS. In 2026 Geothermal Rising Conference: Using the Earth to Save the Earth, Geothermal Resources Council, 2026.
2. **Zhang, Q.**, Dahi Taleghani, A. Simulating Thermal Short-Circuiting in EGS: A Coupled Thermo-Hydro-Mechanical-Wellbore Approach. In ARMA US Rock Mechanics/Geomechanics Symposium, 2026.
3. **Zhang, Q.**, Dahi Taleghani, A. Modeling the Onset of Thermal Short-Circuiting in EGSs with a Coupled THM-Wellbore Approach. In 51st Workshop on Geothermal Reservoir Engineering, Stanford University, Stanford, California, 2026.
4. **Zhang, Q.**, Dahi Taleghani, A. Unlocking Higher Efficiency in CO₂-EGS: The Role of Fluid Flow Control. In 2025 Geothermal Rising Conference: Using the Earth to Save the Earth, Geothermal Resources Council, 2025.
5. **Zhang, Q.**, Dahi Taleghani, A. Operation Strategy to Avoid Thermal Short-circuit in EGSs with Horizontal Wells. In 2024 Geothermal Rising Conference: Using the Earth to Save the Earth, pp. 1116-1127, Geothermal Resources Council, 2024.
6. **Zhang, Q.**, Dahi Taleghani, A. Investigating Fracture Growth After Shut-In. In SPE Annual Technical Conference and Exhibition. OnePetro, 2024.
7. **Zhang, Q.**, Dahi Taleghani, A. Downhole Flow Management to Enhance Efficiency of Fractured Geothermal Systems in Horizontal Wells. In 49th Workshop on Geothermal Reservoir Engineering, Stanford University, Stanford, California, 2024.
8. **Zhang, Q.**, Dahi Taleghani, A. On tunable fracture conductivity to improve heat extraction from fractured geothermal systems. In 48th Workshop on Geothermal Reservoir Engineering, Stanford University, Stanford, California, 2023.
9. **Zhang, Q.**, Dahi Taleghani, A. Variable Fracture Conductivity to Enhance Heat Extraction in EGS. In 2022 Geothermal Rising Conference: Using the Earth to Save the Earth, pp. 542-551, Geothermal Resources Council, 2022.
10. **Zhang, Q.**, Zhu, W., Liu, W., Shi, Y., Yan, J. Study on Characteristics of Water-Gas Flow in Tight Gas Reservoir with High Water Saturation. Gas Injection into Geological Formations and Related Topics, 2020, 321-332.
11. Zhu, W., Huang, K., Sun, Y., **Zhang, Q.** The Description and Modeling of Gas Override in Condensate Gas Reservoir. Gas Injection into Geological Formations and Related Topics, 2020, 333-342.
12. Huang, K., Zhu, W., **Zhang, Q.**, Xia, J., Luo, K. The Study on the Gas Override Phenomenon in Condensate Gas Reservoir. Gas Injection into Geological Formations and Related Topics, 2020, 311-319.

Additional Oral Presentations without Proceedings

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| 1. The 6th Chinese Scientific Data Conference 2019 | Guiyang, China |
| Presentation: Prediction of remaining oil saturation distribution in a water-flooding reservoir based on machine learning methods | 08/2019 |
| 2. The 8th International Acid Gas Injection Symposium (AGIS 2019) | Calgary, Canada |
| Presentation: Study on characteristics of water-gas flow in tight gas reservoir with high water saturation | 09/2019 |
| 3. COMSOL Conference 2019 | Beijing, China |
| Presentation: Numerical simulation of multi-stage fractured horizontal well in a reservoir with moving boundary | 10/2019
Sub-session Chair |
| 4. COMSOL Conference 2018 | Shanghai, China |
| Presentation: Quantitative analysis of effective hydraulic fracture length in exploitation process of tight oil reservoir | 10/2018
Sub-session Chair |

Patents

1. Song, H., **Zhang, Q.**, Li, Z., Du, S., Wang, J. Oil field residual oil saturation distribution prediction method based on machine learning, CN110807544A, Granted October 13, 2020, China.
2. Zhu, W., Wang, Y., Yue, M., Song, Z., Chen, Z., **Zhang, Q.**, Liu, W. Shale gas multi-well group efficient mining interval optimization method, CN109209333B, Granted September 25, 2020, China.

Professional Service and Memberships

Academic Editor, International Journal of Coal Science and Technology 2024 – Present

Coordinator, Repurposing Center for Energy Transition, Penn State 2024 – Present

- Coordinate interdisciplinary research activities on repurposing subsurface infrastructure for geothermal energy and energy storage.

Member, Society of Petroleum Engineers (SPE) 2021 – Present

- SPE Geothermal Technical Section.
- SPE Hydraulic Fracturing Technical Section.

Peer Reviewer, Reviewed manuscripts for more than 30 journals.

- Selected journals: SPE Journal | Energy | Renewable Energy | Fuel | Applied Thermal Engineering | Physics of Fluids | International Journal of Heat and Mass Transfer | Geoenergy Science and Engineering.

Technical Expertise

- **Computational methods:** finite-element, finite-volume, and finite-difference methods, XFEM, cohesive-zone modeling, CFD–DEM, and smoothed particle hydrodynamics.
- **Coupled-process simulation:** thermo-hydro-mechanical modeling, reactive transport, reservoir simulation, and wellbore–reservoir coupling.
- **Programming and scientific computing:** Python, MATLAB, and MRST.
- **Simulation platforms:** COMSOL Multiphysics, ABAQUS, TOUGH2, TOUGHREACT, PHREEQC, CMG, STAR-CCM+, and ANSYS Fluent.
- **Data-driven methods:** physics-informed machine learning, surrogate modeling, and spatiotemporal prediction.

